

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems
COURSE CODE: CSA0501

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.*
(BL2, BL3)

Activity Tool : Case Study (Medium)
Assessment Tool Weightage: 30%

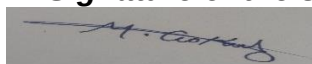
QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Analyze the case: An online shopping platform stores Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price). Identify FDs and normalize to 3NF.	BL2 – Understand	5
2	A hospital maintains Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName). Identify anomalies and normalize to BCNF.	BL3 – Apply	5
3	Case study: A university stores Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade). Analyze FDs and apply 2NF decomposition.	BL3 – Apply	5
4	Given the airline booking case: Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date). Normalize up to BCNF.	BL3 – Apply	5
5	Analyze an HR system schema for a multinational company. Identify all functional and transitive dependencies and normalize to 3NF.	BL2 – Understand	5
6	Case: Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate). Identify MVDs and decompose to 4NF.	BL3 – Apply	5
7	Analyze a telecom billing schema and identify all candidate keys, primary keys, and functional dependencies before normalization.	BL2 – Understand	5

8	A retail inventory system has Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse). Apply complete normalization.	BL3 – Apply	5
9	Study the given poorly normalized schema for a School ERP and propose a fully normalized design up to BCNF with justification.	BL3 – Apply	5
10	Given a movie streaming platform schema, identify join dependencies and decompose to 5NF with lossless-join verification.	BL3 – Apply	5

Code of Conduct:

I M . G O K U L 1 9 2 5 6 5 0 6 7 _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions	Logical solutions with	Basic solutions with weak	Incorrect or no solution provided

		with strong justification	adequate justification	justification	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q1. Online Shopping Platform — Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price) — FDs and 3NF

Assume each row represents one order line item, so OrderID uniquely identifies a line item.

Functional Dependencies:

- OrderID $\rightarrow$ CustID, ProdID, Qty
- CustID $\rightarrow$ CustName
- ProdID $\rightarrow$ ProdName, Price

Candidate key: OrderID (single attribute — {OrderID}⁺ reaches all other attributes through the FDs above).

Transitive dependencies: OrderID $\rightarrow$ CustID $\rightarrow$ CustName, and OrderID $\rightarrow$ ProdID $\rightarrow$ ProdName, Price. These are transitive (CustID and ProdID are non-prime), so the relation violates 3NF.

3NF Decomposition:

Relation	Attributes	Key
ORDERS	OrderID, CustID, ProdID, Qty	OrderID
CUSTOMER	CustID, CustName	CustID
PRODUCT	ProdID, ProdName, Price	ProdID

This removes the redundant repetition of CustName and ProdName/Price across multiple order lines for the same customer/product.

Q2. Hospital — Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName) — Anomalies and BCNF

Functional Dependencies:

- PID $\rightarrow$ PName, DocID, WardNo
- DocID $\rightarrow$ DocName, Specialization
- WardNo $\rightarrow$ WardName

Anomalies in the un-normalized relation:

- Insertion: a new doctor cannot be added to the system until a patient is assigned to them.
- Deletion: discharging the only patient under a doctor deletes that doctor's DocName/Specialization data entirely.
- Update: if a doctor's specialization changes, every row (every patient) under that doctor must be updated, risking inconsistency.

BCNF check: DocID (determinant of DocName, Specialization) is not a superkey of Patient, and WardNo (determinant of WardName) is not a superkey either — both violate BCNF.

BCNF Decomposition:

Relation	Attributes	Key
PATIENT	PID, PName, DocID, WardNo	PID
DOCTOR	DocID, DocName, Specialization	DocID
WARD	WardNo, WardName	WardNo

Every determinant in each resulting relation is now a key, so all three relations satisfy BCNF, and the decomposition is lossless-join.

Q3. University — Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade) — FDs and 2NF

Functional Dependencies:

- $SID \rightarrow SName$
- $CourseID \rightarrow CourseName, Faculty$
- $SID, CourseID \rightarrow Grade$

Candidate key: {SID, CourseID}. SName depends only on SID (part of the key), and CourseName/Faculty depend only on CourseID (part of the key) — these are partial dependencies, violating 2NF.

2NF Decomposition:

Relation	Attributes	Key
STUDENT	SID, SName	SID
COURSE	CourseID, CourseName, Faculty	CourseID
ENROLLMENT	SID, CourseID, Grade	SID, CourseID

Every non-prime attribute in each relation now depends on the whole key of that relation, so all three relations satisfy 2NF.

Q4. Airline Booking — Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date) — Normalize to BCNF

Functional Dependencies:

- $FlightNo \rightarrow DeptCity, ArrCity$ (a flight number fixes its route)
- $PassID \rightarrow PassName$
- $FlightNo, Date, Seat \rightarrow PassID$ (a specific seat on a specific flight instance is held by exactly one passenger)

Candidate key: {FlightNo, Date, Seat} — its closure reaches PassID, PassName, DeptCity, ArrCity, i.e. all attributes.

BCNF check: $FlightNo \rightarrow DeptCity, ArrCity$ — FlightNo alone is not a superkey → violates BCNF. $PassID \rightarrow PassName$ — PassID alone is not a superkey → violates BCNF.

BCNF Decomposition:

Relation	Attributes	Key
FLIGHT	FlightNo, DeptCity, ArrCity	FlightNo
PASSENGER	PassID, PassName	PassID
BOOKING	FlightNo, Date, Seat, PassID	FlightNo, Date, Seat

In BOOKING, the determinant {FlightNo, Date, Seat} is itself the key, so it is in BCNF; FLIGHT and PASSENGER are trivially in BCNF as their determinants are their keys.

Q5. HR System (Multinational Company) — Functional and Transitive Dependencies, Normalize to 3NF

Representative schema: EMPLOYEE(EmpID, EmpName, DeptID, DeptName, CountryCode, CountryName, ManagerID, Salary)

Functional Dependencies:

- EmpID → EmpName, DeptID, ManagerID, Salary
- DeptID → DeptName, CountryCode
- CountryCode → CountryName

Transitive dependencies identified:

- EmpID → DeptID → DeptName (DeptName depends transitively on EmpID)
- EmpID → DeptID → CountryCode → CountryName (a two-step transitive chain)

Since DeptID and CountryCode are non-prime attributes acting as intermediate determinants, these transitive dependencies violate 3NF.

3NF Decomposition:

Relation	Attributes	Key
EMPLOYEE	EmpID, EmpName, DeptID, ManagerID, Salary	EmpID
DEPARTMENT	DeptID, DeptName, CountryCode	DeptID
COUNTRY	CountryCode, CountryName	CountryCode

Each fact (employee, department, country) is now stored exactly once, removing the transitive redundancy that would otherwise repeat DeptName/CountryName for every employee.

Q6. Library — Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate) — MVDs and 4NF

A book can have multiple authors and belong to multiple genres, and these two facts are independent of each other. This causes every (Author, Genre) combination to be repeated for the same book, and independently, every book a member borrows recurs across borrow records.

Functional Dependencies:

- MemberID → MemberName

- $\text{BookID} \rightarrow \text{Title}$

Multi-valued Dependencies:

- $\text{BookID} \twoheadrightarrow \text{Author}$
- $\text{BookID} \twoheadrightarrow \text{Genre}$

Because BookID is not a superkey of the combined (BookID, Author, Genre) grouping, this non-trivial MVD violates 4NF.

4NF Decomposition:

Relation	Attributes
MEMBER	MemberID, MemberName
BOOK	BookID, Title
BOOK_AUTHOR	BookID, Author
BOOK_GENRE	BookID, Genre
BORROW	MemberID, BookID, BorrowDate

Splitting the independent multi-valued facts (Author, Genre) into their own relations removes the redundant repetition while the join of BOOK_AUTHOR and BOOK_GENRE with BOOK still losslessly reconstructs the original book-author-genre associations.

Q7. Telecom Billing Schema — Candidate Keys, Primary Key, and FDs (Before Normalization)

Representative schema: TELECOM_BILL(CustID, CustName, PlanID, PlanName, CallID, CallDate, Duration, Amount)

Functional Dependencies:

- $\text{CallID} \rightarrow \text{CustID}, \text{CallDate}, \text{Duration}, \text{Amount}$ (each call record is uniquely identified by CallID)
- $\text{CustID} \rightarrow \text{CustName}, \text{PlanID}$
- $\text{PlanID} \rightarrow \text{PlanName}$

Candidate key determination:

Closure of {CallID}: $\text{CallID} \rightarrow \text{CustID}, \text{CallDate}, \text{Duration}, \text{Amount}$; then $\text{CustID} \rightarrow \text{CustName}, \text{PlanID}$; then $\text{PlanID} \rightarrow \text{PlanName}$. $\{\text{CallID}\}^+ = \text{all attributes of the relation}$.

Candidate Key = {CallID} → also the Primary Key, since it is the only minimal attribute set whose closure covers all attributes.

(CustID and PlanID are prime-adjacent determinants but cannot serve as keys on their own since they cannot derive CallID, CallDate, Duration, or Amount.)

Q8. Retail Inventory — Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse) — Complete Normalization

Functional Dependencies:

- $\text{PID} \rightarrow \text{PName}, \text{SupplierID}, \text{Category}, \text{Price}, \text{Warehouse}$

- $\text{SupplierID} \rightarrow \text{SupplierName}$

1NF: All attributes already hold atomic values, so the relation satisfies 1NF.

2NF: The only candidate key {PID} is a single attribute, so no partial dependency is possible — 2NF holds.

3NF/BCNF: $\text{SupplierID} \rightarrow \text{SupplierName}$ is a transitive dependency ($\text{PID} \rightarrow \text{SupplierID} \rightarrow \text{SupplierName}$), and SupplierID is not a superkey of Product, so this violates both 3NF and BCNF.

Normalized Design:

Relation	Attributes	Key
PRODUCT	PID, PName, SupplierID, Category, Price, Warehouse	PID
SUPPLIER	SupplierID, SupplierName	SupplierID

Both relations are now in BCNF (each determinant — PID in PRODUCT, SupplierID in SUPPLIER — is a key of its own relation), completing the normalization.

Q9. School ERP — Poorly Normalized Schema — Fully Normalized Design up to BCNF

Given poorly normalized schema: SCHOOL_INFO(StudentID, StudentName, ClassID, ClassName, TeacherID, TeacherName, SubjectID, SubjectName, Marks)

Functional Dependencies:

- $\text{StudentID} \rightarrow \text{StudentName}, \text{ClassID}$
- $\text{ClassID} \rightarrow \text{ClassName}, \text{TeacherID}$
- $\text{TeacherID} \rightarrow \text{TeacherName}$
- $\text{SubjectID} \rightarrow \text{SubjectName}$
- $\text{StudentID}, \text{SubjectID} \rightarrow \text{Marks}$

Issues in the original design: ClassName/TeacherID repeat for every student in a class (redundancy + update anomaly), TeacherName repeats for every class a teacher handles, and deleting the last student of a class removes ClassName/TeacherID data (deletion anomaly).

There is also a transitive chain $\text{StudentID} \rightarrow \text{ClassID} \rightarrow \text{ClassName/TeacherID} \rightarrow \text{TeacherName}$.

Proposed BCNF Design:

Relation	Attributes	Key	Justification
STUDENT	StudentID, StudentName, ClassID	StudentID	Removes transitive dependency on ClassName/TeacherID
CLASS	ClassID, ClassName, TeacherID	ClassID	Removes transitive dependency on TeacherName
TEACHER	TeacherID, TeacherName	TeacherID	Determinant TeacherID is the key — BCNF
SUBJECT	SubjectID, SubjectName	SubjectID	Determinant SubjectID is the key — BCNF

Relation	Attributes	Key	Justification
MARKS	StudentID, SubjectID, Marks	StudentID, SubjectID	Composite key is the only determinant — BCNF

Every determinant in every resulting relation is a candidate key of that relation, so the full design satisfies BCNF, and the decomposition is lossless-join and (for the practical FDs) dependency-preserving.

Q10. Movie Streaming Platform — Join Dependencies and 5NF with Lossless-Join Verification

Representative ternary relationship: STREAM(User, Movie, Device), with the business rule: 'if a user watches a given movie, and the user uses a given device, and that movie is available/watched on that device, then the user watches that movie on that device.' This is a classic 3-way join dependency scenario.

User	Movie	Device
U1	M1	TV
U1	M2	TV
U1	M1	Mobile

Join Dependency: JD(USER_MOVIE, USER_DEVICE, MOVIE_DEVICE) — the natural join of the three binary projections must reconstruct STREAM exactly. Because this ternary fact cannot be losslessly captured by any two of the three binary relations alone, STREAM is not in 5NF.

5NF Decomposition:

Relation	Attributes
USER_MOVIE	User, Movie
USER_DEVICE	User, Device
MOVIE_DEVICE	Movie, Device

Lossless-join verification (Chase/tabular check):

1. Build a tableau with one row per projection (USER_MOVIE, USER_DEVICE, MOVIE_DEVICE), using 'a' for attributes present in that projection and subscripted 'b' symbols for attributes missing from it.
2. Apply the join-dependency rule: whenever two rows agree on the attribute(s) shared between two of the projections, propagate the matching values (replace a 'b' with an 'a' where forced by an actual data row, following the same equating logic as the FD-based chase).
3. After propagation, at least one row becomes entirely 'a' (matching STREAM exactly) whenever the underlying data actually satisfies the 3-way consistency rule shown above — confirming JD(USER_MOVIE, USER_DEVICE, MOVIE_DEVICE) holds and the decomposition is lossless-join.

This 5NF decomposition removes the redundancy caused by the join dependency: each pairwise fact (which movies a user watches, which devices a user owns, which movies are available on which device) is stored exactly once, and the full three-way relationship is recoverable only by joining all three relations together